

PURCHASE DESCRIPTION

1. Statement of Need and Purpose:

The Antibody Diversity Section at the National Institute on Aging (NIA) is investigating the hypothesis that patients with Alzheimer's disease exhibit increased immune cell infiltration into the cerebrospinal fluid (CSF). To support this research, the laboratory will perform single-cell RNA sequencing (scRNA-seq) on cellular pellets from approximately 60 human CSF samples. This requirement is for reagents necessary to partition cells into single-cell Gel Beads-in-Emulsion (GEMs) and generate barcoded cDNA libraries for downstream RNA sequencing analysis. These reagents will enable identification of immune cell populations and characterization of transcriptional differences between healthy and disease states.

2. Background Information and Objective:

During Alzheimer's disease progression, immune cells accumulate in the CSF. Previous studies conducted by the laboratory have identified B cells within CSF and characterized antibody production. The current effort expands this work to additional patient samples using single-cell RNA sequencing. The requested reagents must be compatible with existing laboratory instrumentation and workflows to ensure consistency with previously generated datasets and maintain continuity in experimental methods.

3. Generic Name of Product:

Single-cell RNA sequencing (scRNA-seq) reagents for cell partitioning, barcoding, and library preparation.

4. Purchase Description:

Single-cell RNA sequencing reagents compatible with the 10x Genomics Chromium platform, including:

- Library Construction Kit, 16 reactions (Part No. 1000190)
- GEM-X Universal 5' Gene Expression v3, 16 samples
- Associated shipping and handling

Salient characteristics:

The required reagents must meet the following minimum technical characteristics:

- Compatibility with the 10x Genomics Chromium Controller system currently in use at NIA
- Capability to partition individual cells into Gel Beads-in-Emulsion (GEMs)
- Support for 5' gene expression profiling workflows
- Integrated single-cell barcoding for downstream RNA sequencing
- Validated performance for human-derived biological samples, including CSF

- Ability to generate high-quality cDNA libraries suitable for next-generation sequencing

6. Quantity:

Quantity 2: Library Construction Kit, 16 reactions (Part No. 1000190)

Quantity 1: GEM-X Universal 5' Gene Expression v3, 16 samples

7. Delivery Date: Thirty days after the award of the contract.

Deliver to:

Robert W. Maul, PhD.

Biomedical Research Center, Rm 06C009

251 Bayview Blvd.

Baltimore, MD 21224

8. Period of Performance: Not applicable. This is a supply purchase. Standard manufacturer warranty provisions shall apply.

9. Option(s): N/A